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AGRICULTURE

Corn Leafhopper

Dalbulus maidis

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Description of the Pest

The adult corn leafhopper is light tan in color and about 1/8 of an inch long. Its most distinguishing feature is [two dark spots](#) located between the eyes, which are visible using a 10X hand lens. The [nymphs](#) have no wings and are green to tan in color. They run rapidly across the under surface of the leaf when disturbed and may move from side to side and even backwards. Both adults and nymphs like to feed [inside the whorl](#), particularly in young corn. Later, as the plants grow, they move out onto the underside of the leaves.

Corn leafhoppers overwinter as adults in the southern San Joaquin Valley. They prefer areas where they can find shelter such as [grassy](#).

[vegetation](#) along waterways, ditch banks, and fence lines. [Alfalfa fields](#) are also a preferred overwintering site. The adults do not feed on alfalfa; they simply take up residence in the crowns where they spend the winter. As temperatures warm in spring, adult leafhoppers become active and fly around searching for corn. As soon as corn emerges, the adults move from their overwintering sites into the newly planted fields.

Damage

Corn leafhopper causes damage in two ways. First, leafhoppers directly feed on the plant, sucking out juices. Heavy populations can cause the leaves to dry; also, both the adults and nymphs produce sticky honeydew while they feed, which gets on the corn leaves. Black sooty mold frequently grows on the honeydew, reducing the photosynthetic capacity of the plant. Secondly, and more importantly, the leafhoppers transmit a pathogen called *Spiroplasma kunkelii*, a bacteria-like organism that causes the disease corn stunt. Corn stunt is much more debilitating to the plants than the direct feeding damage caused by the leafhopper. The pathogen responsible for corn stunt overwinters within the adult leafhopper, so leafhoppers emerging from overwintering in early spring can be infective, as can later generations. Corn stunt causes plants to be [stunted](#) and can cause significant yield losses. For more information on this disease, see the section on corn stunt.

Until recently corn leafhopper was only a problem on late-planted corn (planted after July 1). Over the past few years, however, it has become a serious pest on early planted corn as well, so leafhoppers may be found on corn as early as March and April. Corn leafhoppers damage silage corn, grain corn, and sweet corn.

Distribution

In the Central Valley, the corn leafhopper has been identified from Kern, Kings, Tulare, Fresno, Madera, Merced, Stanislaus, San Joaquin, Sacramento, Solano, and Yolo counties. It has also been reported from Riverside and Los Angeles counties. Corn stunt disease has been identified in Kern, Kings, Tulare and Fresno counties. The spiroplasma that causes the disease has been isolated from leafhoppers collected in Sacramento County, but the corn stunt disease has not yet been found.

Management

Early planting and maintaining a corn-free period over the winter months are key strategies in avoiding damage from the corn leafhopper and the incidence of corn stunt disease. In sweet corn, the use of reflective mulches may be a feasible management option. Chemicals are not effective at reducing the spread of the corn stunt spiroplasma by the leafhopper.

Biological Control

While a parasite has been reported from Mexico and South America, it has not been found in California. Large populations of minute pirate bugs, green lacewings, and assassin bugs are found associated with corn leafhopper populations, however, these common predators appear to have little effect on leafhopper populations.

Cultural Control

Leafhopper infestations and corn stunt disease become more severe as the growing season progresses. Planting as early as possible may help to lower the infestation rate of both, however, it will not eliminate the problem. Reflective plastic mulch has been shown to repel the adult leafhoppers and reduce the incidence of corn stunt disease in sweet corn. While this strategy is too expensive for use in silage or grain corn fields, it may be practical for sweet corn production. Growers are urged to control volunteer corn plants that

emerge following harvest. This includes volunteer corn in alfalfa fields and in winter forage crops planted after corn harvest. The volunteer corn plants are often present until killed by frost. In mild winters, these plants may survive the entire winter, and live adults and nymphs can be found deep within the whorl. Because corn is the only host for the corn leafhopper in California, these volunteer plants provide a winter host for the leafhopper and the spiroplasma, allowing them to survive. It is highly recommended that all corn be harvested no later than October 31 to provide the longest corn-free period possible.

In no-till systems, when corn is planted as a double crop following wheat or barley, the straw mulch has been shown experimentally to reduce the incidence of both the corn leafhopper and corn stunt disease. Additional research is needed, however, before this strategy can be proposed as a general management tool.

Organically Acceptable Methods

Planting over reflective mulch for management in sweet corn as well as the other management strategies listed under Cultural Controls are acceptable for use in organically certified corn.

Treatment Decisions

Treatment thresholds have not been established and insecticide treatments are not recommended. It is important to remember when dealing with a disease vector, that threshold values would be extremely low. A single, infected vector may inoculate many plants during its lifetime. While there are insecticides registered to control corn leafhopper on corn, they are not effective in preventing the incidence of corn stunt disease. Seed treatments with systemic insecticides have been shown to reduce corn leafhopper populations, but incidence of corn stunt disease has only been negligibly reduced. Reinfestations occur rapidly, particularly when other fields of corn in the area are being harvested, and leafhoppers are searching for other corn plants.



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